How to create an Engine

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1 REQUIREMENTS 2

2 CREATE THE FOLDER STRUCTURE 3

3 CREATE THE ENGINE GEOMETRY 4

4 CREATE THE ENGINE TEXTURE 6

5 CREATE THE ENGINE BLUEPRINT 7

5.1 Create the Engine Blueprint 7

5.2 Create the Bogie Blueprints 7

5.3 Create the Simulation Blueprint 8

5.3.1 Create the Data Graphs 9

5.3.2 Continued in the Simulation Blueprint 10

5.3.3 Create the Simulation Script 11

5.3.4 Continued in the Simulation Blueprint 11

5.4 Continued in the Engine Blueprint 11

5.5 Create the Cab Camera position 11

5.6 Continued in the Engine Blueprint 12

5.7 Create the Train Controls 12

5.8 Create the Script Component 14

5.8.1 Continued in the Engine Blueprint 14

5.9 Create the fuel Tank 14

6 ADD THE ENGINE INTO A ROUTE 15

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1 Requirements

- Rail Simulator + Upgrade Mk1
- RSDevTools
- Art Plugins installed (available from www.railsimulator.com)
- 3D Object Creation Software (3D Studio Max7 is used in this tutorial)
- 2D Texture Creation Software (Photoshop is used in this tutorial)
- At least basic knowledge of modelling, texturing and mapping

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2 Create the folder structure

For this we are going to pretend our engine is a diesel locomotive so we will be setting up the folder structure for diesel engines. Examples of folder structures can be found in the provided developer tools in; *C:|Program Files|Rail Simulator|Source|Developer|Addon*

Remember that the *Developer* name can be any you wish for yourself, and the *Addon* name can be any product you wish to call it. For default Rail Simulator, the developer is *Kuju* and the Product is *RailSimulator*. For the Class 08 Shunter, the developer was *RSDL* and the product was again *RailSimulator*. We could have chosen *DevTeam* and *Class08* if we wanted.

For this tutorial I am going to call the Provider *RSadam* and the Product *HowTo*. You can use what ever names you wish.

So create your new folder structure like this;

C:\Program Files\Rail Simulator\Source*RSadam**HowTo***RailVehicles\Diesel**

- Inside the **Diesel** folder, create another called **SimpleEngine**
- Inside the **SimpleEngine** folder we need a structure to keep everything neat
- Create a folder named **Engine**. This is where we will save any max, xml and igs files related to our model
- Inside the **Engine** folder make another folder called **Textures**
- Inside the new **Textures** folder save any psd, tga and ace files related to our model

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3 Create the Engine Geometry

1. Open up 3D Studio Max
2. Create a box in the top view, 18m x 2.5m x 2.5m
3. Name the box: ***1_0500_engine*** (Naming conventions are explained in devdoc 4.03)
4. Still in the top view, make sure ***select and move*** is highlighted, press ***F12*** (or click on ***tools*** *>>* ***transform type*** in on the menu bar)
5. In the X and Y fields in the absolute world column press ***0*** so that the box is centred on the world axis. In the Z field type ***1.25*** to raise the body off the ground.
6. Right click the box and click ***convert to: editable poly***
7. Click the modify tab then select ***polygon***
8. Left click and draw a selection box around the shape in the top screen
9. In the modify list select ***UVW map***
10. Change the mapping type to ***box***
11. In the top screen right click on the shape and ***convert to edit poly***

Now we have the main body, we need some bogies

12. Create a box in the top view, 7.5m x 2m x 0.75m
13. Name the box: ***1_0500_bo01*** (Naming conventions are explained in devdoc 4.03)
14. Still in the top view, make sure ***select and move*** is highlighted, press ***F12*** (or click on ***tools*** *>>* ***transform type*** in on the menu bar)
15. Leave the X field at ***0***. In the World column Y field type ***5.585*** and in the Z field type ***0.3***.
16. Right click the box and click ***convert to: editable poly***
17. Click the modify tab then select ***polygon***
18. Left click and draw a selection box around the shape in the top screen
19. In the modify list select ***UVW map***
20. Change the mapping type to ***box***
21. In the top screen right click on the shape and ***convert to edit poly***
22. We now have one bogie in place. Now for the other.
23. Create a third box in the top view, 7.5m x 2m x 0.75m
24. Name the box: ***1_0500_bo02*** (Naming conventions are explained in devdoc 4.03)
25. Still in the top view, make sure ***select and move*** is highlighted, press ***F12*** (or click on ***tools*** *>>* ***transform type*** in on the menu bar)
26. Leave the X field at ***0***. In the World column Y field type -***5.585*** and in the Z field type ***0.3***.
27. Right click the box and click ***convert to: editable poly***
28. Click the modify tab then select ***polygon***
29. Left click and draw a selection box around the shape in the top screen
30. In the modify list select ***UVW map***
31. Change the mapping type to ***box***
32. In the top screen right click on the shape and ***convert to edit poly***

We now have our two bogies in place. So let's add some wheels

33. This time in the left view, create a cylinder, radius ***0.56m*** x height ***1.5m***
34. Name the cylinder: ***1_0500_bo01wh01***
35. Select the top view pane and click the Hierarchy tab.
36. Make sure the Pivot button is active, in the Adjust Pivot section, press ***Affect Pivot Only*** and then ***Centre to Object*** and ***Align to World***

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37. Now in the left view, make sure ***select and move*** is highlighted, press ***F12*** (or click on ***tools >> transform type*** in on the menu bar)
38. Make the X field ***0***. In the World column Y field type ***7.802*** and in the Z field type ***0.555***.
39. Right click the box and click ***convert to: editable poly***
40. Click the modify tab then select ***polygon***
41. Left click and draw a selection box around the shape in the top screen
42. In the modify list select ***UVW map***
43. Change the mapping type to ***cylinder***
44. In the top screen right click on the shape and ***convert to edit poly***
45. We now need the other 5 wheels. Repeat steps 35 to 46 with the following details:

1_0500_bo01wh01	X: ***0***	Y: ***7.802***	Z: ***0.555***
1_0500_bo01wh02	X: ***0***	Y: ***5.585***	Z: ***0.555***
1_0500_bo01wh03	X: ***0***	Y: ***3.379***	Z: ***0.555***
1_0500_bo02wh01	X: ***0***	Y: ***-7.802***	Z: ***0.555***
1_0500_bo02wh02	X: ***0***	Y: ***-5.585***	Z: ***0.555***
1_0500_bo02wh03	X: ***0***	Y: ***-3.379***	Z: ***0.555***

46. Lastly we need to connect the wheels to each bogie.
47. In the left view, select ***1_0500_bo01wh01*** and click the ***Select and Link*** tool
48. Click on the wheel and drag the mouse onto ***1_0500_bo01***. This will link the wheel to the bogie
49. Do the same for the other two wheels under ***1_0500_bo01*** and then repeat for the wheels under ***1_0500_bo02***

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4 Create the Engine Texture

1. Open up Adobe Photoshop CS
2. Click on ***File >> New*** and type in ***256*** in the width and ***512*** in the height column
3. Select a brush and colour and draw a simple pattern in the image
4. Click on ***File >> Save as***
5. Select the .***ACE*** in the type box and save the file in:
   \Source*RSadam**HowTo*\RailVehicles\Diesel\SimpleEngine\Engine\Textures

Note - as it is a 24 bit texture there is no need to change any of the fields, just click Save

6. Return to 3D Studio Max.
7. Select the ***Material Editor*** (the 4 coloured spheres icon on the menu bar)
8. Click on the ***Standard*** tab and select ***Kuju Material***, then click ***OK***
9. Select ***TexDiff.fx*** shader from the ***Shader Name*** pull down menu

Note – shaders are explained in the dev doc 4.02 – shaders

10. Click on the texture slot which says ***None***
11. Find the texture in your texture folder
    \Source*RSadam**HowTo*\RailVehicles\Diesel\SimpleEngine\Engine\Textures and click ***open***
12. Assign the material to each shape we have just created by clicking on the ***assign to*** icon which looks like a blue sphere with an arrow pointing to a white cube.
13. Click on the chequered blue and white cube icon, this shows the material in the view port.
14. Right click on the words ***Smooth and Highlights*** located in the top left corner of the view port screen.

Note – you should be able to see your simple texture on your engine.

15. Select ***File >> Export***
16. Select ***Kuju Intermediate Geometry (IGS,*.IGL)*** from the list
17. Type in the name field: ***SimpleEngine*** and click save (make sure you are saving the file in ..\Source*RSadam**HowTo*\RailVehicles\Diesel\SimpleEngine\Engine)
18. Ensure the ***Export with Name Spaces*** box is ticked
19. Click ***Export*** and then close 3D Studio Max

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5 Create the Engine Blueprint

5.1 Create the Engine Blueprint

1. Open up the ***Asset Editor*** from the shortcut you have create

Note – the Asset Editor requires the Rail Simulator Disc in the Drive

2. In the Blueprint Editor window, find your ***SimpleEngine*** in the folder structure
3. Right click the ***Engine*** folder name and select ***New Blueprint*** from the list
4. Select ***Engine Blueprint*** from the menu that appears and then click ***OK***
5. Right click on the Engine blueprint000 file in the folder tree
6. Rename the file ***SimpleEngine***
7. In the top Name field type ***SimpleEngine***
8. In the Browse Information section type ***SimpleEngine*** in the language fields
9. In the Mass field, type ***50***. This is the weight in Tons
10. For the front pivot X, input ***9.2*** and the back ***-9.2***
11. For Collision Centre Y, we need the centre of the main body sat on the wheels. This is calculated by taking the full height of the vehicle (3.75m) subtracting the value below the main body (1.25m), dividing the remainder my 2 (2.5m) and adding back the height below the main body = ***2.5m***
12. For the Collision width, type ***2.5***
13. For the Collision height, type ***2.5***
14. For the Collision length, type ***18m***

Note - this is less than the pivot points at either end. This is so the coupling points are not inside the hit box.

15. Expand the Bogey section and click *Insert First* two times
16. This will be where the bogie details are referenced. So we need to create some bogie blueprints
17. Click the ***Save*** button at the top of the Engine Blueprint

5.2 Create the Bogie Blueprints

1. On the Folder tree, right click the ***Engine*** folder name
2. Select ***New Blueprint*** from the menu and scroll to ***Bogey Blueprint*** and click ***OK***
3. For Wheel Radius, type ***0.56***
4. For the Wheel gauge, type ***1.435***
5. For Geometry, type ***bo01***
6. Insert three axles in the Axle section
7. Add the wheel radius as above
8. The Horizontal offset is how far from the middle of the bogie each axle is located
9. For the first axle, type ***2.217*** for the Horizontal Offset
10. Add the Node ID of ***bo01wh01***
11. For the second axle, the Horizontal offset is **0**
12. Add the Node ID of ***bo01wh02***
13. For the third axle, type ***-2.206*** for the Horizontal Offset
14. Add the Node ID of ***bo01wh03***
18. Finally add a Pivot Offset and leave the fields as 0
19. Right Click on ***Bogey blueprint001*** in the folder tree and rename this to ***Front Bogie***

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20. Click ***Save***
21. Repeat steps 15 >> 28 with a new Geometry name of ***bo02*** as well as in front of each wheel node ID
22. Rename the blueprint to ***Back Bogie*** and Click ***Save***

23. Return to the Engine Blueprint
24. Scroll down to the Bogey section and expand the two entries
25. In the first entry, type ***5.585*** for the Bogey Pivot X
26. Fill in the Provider and Product fields (***RSadam*** & ***HowTo***)
27. Right Click on the Front Bogie blueprint in the folder tree
28. Select ***Copy Filename*** then paste this into the Blueprint ID field (RailVehicles\Diesel\SimpleEngine\Engine\Front Bogie.xml)
29. In the second entry, type -5.585 for the Bogey Pivot X
30. Fill in the Provider and Product fields (***RSadam*** & ***HowTo***)
31. Right Click on the Back Bogie blueprint in the folder tree
32. Select ***Copy Filename*** then paste this into the Blueprint ID field (RailVehicles\Diesel\SimpleEngine\Engine\Back Bogie.xml)
33. Minimise the Bogey section of the Engine Blueprint
34. In the Engine Type field, choose ***Diesel***
35. Type ***SimpleEngine*** in the Engine Name field
36. You can now minimise the Browser and Rail Vehicle Component sections
37. In the Render Component section, scroll down to the Geometry ID field
38. Right Click on the ***SimpleEngine.igs*** file in the folder tree (This is the file with an icon that looks like a blue cube with an orange arrow in it)
39. Select Copy File name, and then paste this into the Geometry and Collision ID fields (RailVehicles\Diesel\SimpleEngine\Engine\SimpleEngine.IGS)
40. Set the Detail Level Generation Range to ***10***, replacing 1
41. In the Engine Simulation Container section, fill in the Provider and Product fields (*RSadam* and *HowTo*)
42. For the Blueprint ID field we will need a Simulation Blueprint
43. Click the ***Save*** button at the top of the Engine Blueprint

5.3 Create the Simulation Blueprint

1. On the Folder tree, right click the ***SimpleEngine*** folder name
2. Select ***New Folder*** and name it ***Simulation***
3. Right click on this new folder and select **New Blueprint**
4. This time scroll down to ***Engine Sim Blueprint*** and Click ***OK***
5. Name this blueprint ***SimpleEngine Simulation***
6. Fill in the Name and Language fields with ***SimpleEngine Simulation***
7. Expand the Engine Sim Component Sub System and click ***Insert First***
8. Select Diesel Electric Sub System Blueprint for the menu that appears

Note – please see the Engine Simulation document for details on prototypical setup of Simulation data

9. Expand the new simulation entry
10. Expand the Train Brake Assembly section and click the ***Insert First*** button
11. Expand to the Brake Type section and click the ***Insert First*** button
12. Choose ***Train Dual Brake Data*** from the menu that opens
13. Expand all the following sections
14. Change the equipment type to ***Vacuum Single Pipe***
15. Open the Vacuum Brake ***Control*** section
16. Click the ***Insert First*** button

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17. Expand the ***Brake Control*** section
18. Change the Cab Control Name to ***TrainBrakeControl***
19. You can now Minimise the ***Vacuum Brake*** section
20. Expand ***Control*** and click the ***Insert First*** button
21. Change the Cab Control Name to ***TrainBrakeControl***
22. You can now minimise the ***Control*** Section
23. Expand the Emergency Braking section and click the ***Insert First*** button
24. Expand the Auxiliary Braking section and click the ***Insert First*** button
25. You can now minimise the Train Dual Brake section
26. Change the Startup time to 0.5
27. Change the Shutdown time to 0.5

Note - This is simply for test purposes to speed up usage

28. For the Tractive Force v Speed field we will need to create a graph of appropriate values.
29. Click the ***Save*** button at the top of the screen

5.3.1 Create the Data Graphs

Note – for full details on the data graphs used in Rail Simulator, please see the Simulation documents

1. Open Notepad and type the following:
   0,245
   10,245
   20,178
   30,120
   40,89
   50,72
   60,62
   70,53
   80,47
   90,36
   95,33
   100,29

2. Save this file in our ***Simulation*** folder as a .txt file
   Source*RSadam**HowTo*\RailVehicles\Diesel\SimpleEngine\Simulation
3. In the Simulation folder, rename this file ***TractiveEffortVsSpeed.csv***

Note – CSV stands for Comma Separated Variable. So in this format our data is converted into two columns of Tractive effort values compared to the speed of the train

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4. Open Notepad again and type the following:
   0,0
   1,2.4
   10,24.4
   20,48.9
   30,73.2
   40,97.8
   50,122.2
   60,146.7
   70,171.1
   80,195.6
   90,220.1
   100,244

5. Save this file in our ***Simulation*** folder as a .txt file
   Source*RSadam**HowTo*\RailVehicles\Diesel\SimpleEngine\Simulation
6. In the Simulation folder, rename this file ***TractiveEffortVsThrottle.csv***

7. Open Notepad again and type the following:
   0,0
   1,2.4
   10,24.4
   20,48.9
   30,73.2
   40,97.8
   50,122.2
   60,146.7
   70,171.1
   80,195.6
   90,220.1
   100,244

8. Save this file in our ***Simulation*** folder as a .txt file
   Source*RSadam**HowTo*\RailVehicles\Diesel\SimpleEngine\Simulation
9. In the Simulation folder, rename this file ***TractiveEffortVsFieldDivert.csv***

10. Return to our Simulation Blueprint
11. In the folder tree window, minimise the simulation folder and then expand it again. Our CSV files now appear in the folder

5.3.2 Continued in the Simulation Blueprint

1. Right click on the ***TractiveEffortVsSpeed.csv*** file and select ***Copy Filename***
2. Paste this into the ***TractiveEffortVsSpeed*** field
3. Right click on the ***TractiveEffortVsThrottle.csv*** file and select ***Copy Filename***
4. Paste this into the ***TractiveEffortVsThrottle*** field
5. Type ***140*** for the Main Res Max Air Pressure
6. Minimise the Air System Section
7. In the Overload current, type ***1500***
8. Right click on the ***TractiveEffortVsFieldDivert.csv*** file and select ***Copy Filename***
9. Paste this into the ***TractiveEffortVsFieldDivert*** field
10. Expand the Power Unit section and click the ***Insert First*** button
11. Type ***200*** for the Idle RPM value

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12. Type ***750*** for the Max RPM value
13. Type ***50*** for the Change per Second value
14. Type ***2000*** for the Power at Rail Value
15. You can now minimise the entire Engine Sim Component section

5.3.3 Create the Simulation Script

1. Open Notepad again and type the following:
   function Setup ()
   end
   function Update (inteval)
   end
2. Save this file in our ***Simulation*** folder as a .txt file
   \Source*RSadam**HowTo*\RailVehicles\Diesel\SimpleEngine\Simulation
3. In the Simulation folder, rename this file ***SimpleEngine Simulation.lua***

5.3.4 Continued in the Simulation Blueprint

1. Right click on the ***SimpleEngine Simulation.lua*** file and select ***Copy Filename***
2. Paste this into the ***Name*** field of the Script Component
3. Click the ***Save*** button at the top of the screen

Your Engine Sim Blueprint is now complete

5.4 Continued in the Engine Blueprint

1. Minimise Browse Information
2. Minimise RailVehicle Component
3. Minimise Render component
4. In the Engine Simulation Container, fill in the Provider and Product fields (***RSadam*** and ***HowTo***)
5. Right click on the Simulation Blueprint we have just created. This is the file with a blue L and orange circle on it
6. Select ***Copy Filename*** from the list
7. Paste this into the ***Blueprint ID*** field
   (RailVehicles\Diesel\SimpleEngine\Simulation\SimpleEngine Simulation.xml)

5.5 Create the Cab Camera position

1. Fill in the Provider and Product fields for the Interior Camera section (***RSadam*** and ***HowTo***)
2. Click the ***Save*** button at the top of the screen
3. On the Folder tree, right click the ***Engine*** folder name
4. Select ***New Blueprint*** from the menu and scroll to ***Cab Camera Blueprint*** and click ***OK***
5. Enter the name as ***SimpleEngineCabCam***
6. Rename the blueprint to the same name
7. Change the Cam Type value to ***1***
8. Enter ***2.5*** in the Y value *(this is the height above the rails)*
9. Enter ***8.2*** in the Z value *(this is the distance from the centre of the vehicle)*
10. In the Camera Direction section, change Z to ***1***
11. For the Vertical and Horizontal freedom fields, enter ***1080***
12. You now have a camera position blueprint

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5.6 Continued in the Engine Blueprint

1. Minimise Browse Information
2. Minimise RailVehicle Component
3. Minimise Render component
4. Minimise the Engine Simulation Container
5. In the Interior Camera section, we now want to point the blueprint ID to our new blueprint
6. Right click on the SimpleEngineCabCam we have just created. This is the file with a blue L and orange circle on it
7. Select ***Copy Filename*** from the list
8. Paste this into the ***Blueprint ID*** field
   (RailVehicles\Diesel\SimpleEngine\Engine\SimpleEngineCabCam.xml)

9. You can now minimise the Interior Interface section
10. Minimise the Carriage Interior Interface section

5.7 Create the Train Controls

1. Click the ***Insert First*** button under the Control Values section

*Note – please see the **Cab Control** document for details on prototypical setup of controls. For this example we will only setup basic Simple Controls*

2. Expand the new Control Value and name it ***Reverser***
3. Type ***1*** in the Default value field
4. Type ***-1*** in the Minimum Value field
5. Expand the ***Interface Element*** and Click the ***Insert First*** button
6. Scroll through the list and select ***Interior Notched Lever*** and click ***OK***
7. Expand the Interior Notched Lever section
8. Type ***Reverser_Intermediate*** for the Element name
9. Change the difficulty type to ***Intermediate***
10. Change the number of Notches to ***3***
11. Minimise the Interface Element section
12. Click the tall button on the left with the **>** symbol on it

13. Expand the new Control Value and name it ***SimpleChangeDirection***
14. Type ***1*** in the Default value field
15. Type ***-1*** in the Minimum Value field
16. Expand the ***Interface Element*** and Click the ***Insert First*** button
17. Scroll through the list and select ***Interior Lever*** and click ***OK***
18. Expand the Interior Lever section
19. Type ***SimpleChangeDirection*** for the Element name
20. Change the difficulty type to ***Stop go only***
21. Minimise the Interface Element section
22. Click the tall button on the left with the **>** symbol on it

23. Expand the new Control Value that appears below
24. For the Control Name, type ***Regulator***
25. Expand the Interface Element section and click the Insert First button
26. Scroll through the list and select ***Interior Lever*** and click ***OK***
27. Expand the Interior Lever section
28. Type ***Regulator_Intermediate*** for the Element name
29. Change the difficulty type to ***Intermediate***

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30. Minimise the Interface Element section
31. Click the tall button on the left with the **>** symbol on it

32. Expand the new Control Value that appears below
33. For the Control Name, type ***SimpleThrottle***
34. Type ***0.25*** in the Default value Field
35. Expand the Interface Element section and click the Insert First button
36. Scroll through the list and select ***Interior Lever*** and click ***OK***
37. Expand the Interior Lever section
38. Type ***SimpleThrottle_StopGoOnly*** for the Element name
39. Change the difficulty type to ***Stop go only***
40. Minimise the Interface Element section
41. Click the tall button on the left with the **>** symbol on it

42. Expand the new Control Value that appears below
43. For the Control Name, type ***EngineStart***
44. Expand the Interface Element section and click the Insert First button
45. Scroll through the list and select ***Interior Push Button*** and click ***OK***
46. Expand the Interior Push Button section
47. Type ***EngineStat*** for the Element name
48. Change the difficulty type to ***Intermediate***
49. Change the Movement type to ***Push Button***
50. Set the Auto Release value as ***True***
51. Minimise the Interface Element section
52. Click the tall button on the left with the **>** symbol on it

53. Expand the new Control Value that appears below
54. For the Control Name, type ***TrainBrakeControl***
55. Type ***0.25*** for the Default Value
56. Expand the Interface Element section and click the Insert First button
57. Scroll through the list and select ***Irregular Notched Lever*** and click ***OK***
58. Expand the Irregular Notched Lever section
59. Type ***TrainBrakeControl_Intermediate*** for the Element name
60. Change the difficulty type to ***Intermediate***
61. Set the Rest in notched position to ***False***
62. Expand the Notches section and click ***Insert First*** until there are ***5 notches***

Note – There needn't be 5 notches, this is just an example

63. Expand the first notch and type ***Release*** as the Identifier
64. Expand the second notch and type ***GraduatedSelfLapLimitedHolding*** as the Identifier
65. Change the value to ***0.3***
66. Expand the third notch and type ***Suppression*** as the Identifier
67. Change the value to ***0.85***
68. Expand the fourth notch and type ***ContinuousService*** as the Identifier
69. Change the value to ***0.9***
70. Expand the last notch and type ***Emergency*** as the Identifier
71. Change the value to ***0.95***
72. Minimise the all the Control Value sections you have created
73. Click the ***Save*** button at the top of the screen

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5.8 Create the Script Component

1. Open Notepad again and type the following:
```
function OnControlValueChange ( name, index, value )
        if Call( "*:ControlExists", name, index ) then
                Call( "*:SetControlValue", name, index, value );
        end
end
```
2. Save this file in our ***Engine*** folder as a .txt file
   \Source*RSadam**HowTo*\RailVehicles\Diesel\SimpleEngine\Engine
3. In the Engine folder, rename this file ***SimpleEngine_Script.lua***
4. Return to our Engine Blueprint
5. In the folder tree window, minimise the Engine folder and then expand it again. Our LUA file now appears in the folder

5.8.1 Continued in the Engine Blueprint

1. Minimise Browse Information
2. Minimise RailVehicle Component
3. Minimise Render component
4. Minimise Engine Simulation Container
5. Minimise Control Container Component
6. Minimise Container Component
7. Right click the ***SimpleEngine_Script.lua*** file and select ***Copy Filename***
8. Paste this into the ***Script Component Name*** field in the Engine Blueprint
9. Minimise Script Component

5.9 Create the fuel Tank

1. Expand the Cargo Def… section
2. Click the ***Insert First*** button
3. Choose ***Bulk Cargo*** from the list and click ***OK***
4. Expand the Bulk Cargo def section
5. Type ***1*** for the ***Width***, ***Height*** and ***Depth*** of the Trigger Box
6. Type ***500*** for the Capacity
7. Select ***Diesel Fuel*** from the Bulk Freight type list
8. Click the ***Save*** button at the top of the screen

Your Engine Blueprint is now complete
Click ***Export*** at the top of the screen to load your engine into Rail Simulator

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6 Add the Engine into a route

1. Ensure you are running in Simple Control Mode
2. Select your route of choice and click the ***Free Play***
3. Click the ***World Editor*** button on the bottom tab (looks like a globe)
4. Click the Train icon in the top left pane. This will open the Scenario Editor.
5. Open the middle tab on the left hand side and click the ***ObjectSet Filter*** (looks like a blue cube with an orange arrow)
6. 'Pin open' the tab on the right hand side and click on the grey arrow next to the name ***Developer***
7. Select our provider from the list (***RSadam***)
8. Put a tick in the box next to our product (***HowTo***)

Note – all objects under that provider and product are now added to the object browser

9. In the middle tab on the left hand side click on the ***Engines*** icon (small cartoon of a High Speed Train power car)
10. Scroll down the list in the middle tab until you see ***SimpleEngine***
11. Select it by left clicking and then moving the mouse into the 3D environment
12. Click the train onto some track
13. Double right click to exit the placement tool
14. Click the ***Driver*** icon in the top window (small profile of a drivers head)
15. Click this onto your new train
16. Press the ***Play*** button to return to drive mode

You now have a new engine for use in Rail Simulator.

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